

Section 1

1. The raw inputs are accX, accY, and accZ.
2. NN classifier features:
 - a. Type: Spectral features
 - b. Number of input features: 33
 - c. 5 feature names: accX RMS, accX Peak 1 Height, accY Peak 3 Height, accX Peak 1 Freq, accY RMS
3. Outputs 2 classes: "nominal" (for when the fan is operating normally) and "off" (for when the fan is off)
4. The anomaly detector uses spectral features as input, and there are a total of 4 used features:
 - a. accX RMS
 - b. accX Peak 1 Height
 - c. accY Spectral Power 2.0 - 5.0 Hz
 - d. accZ Spectral Power 2.0 - 5.0 Hz
- 5.

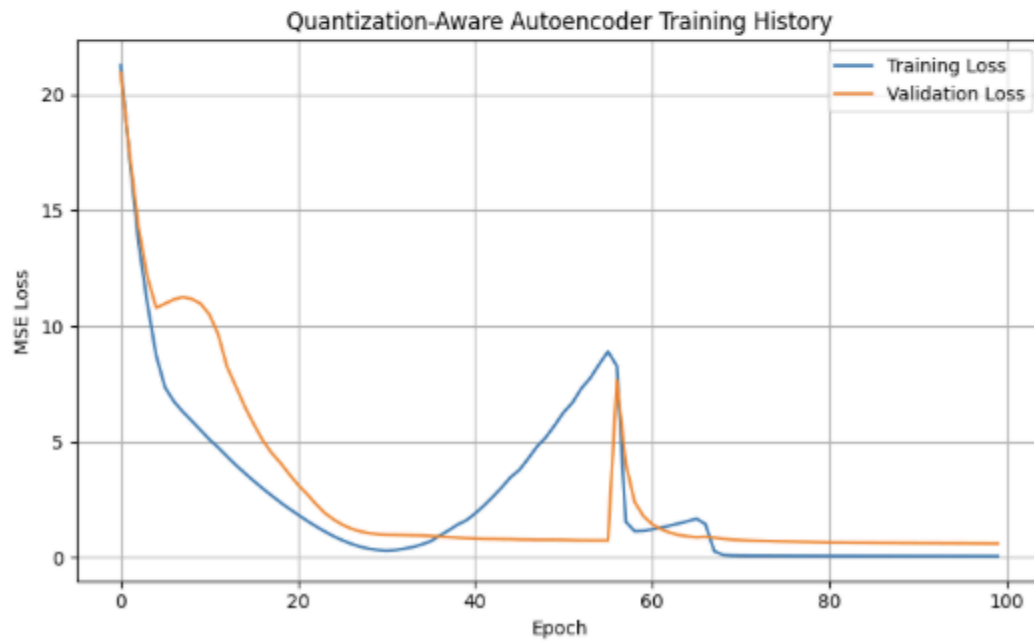
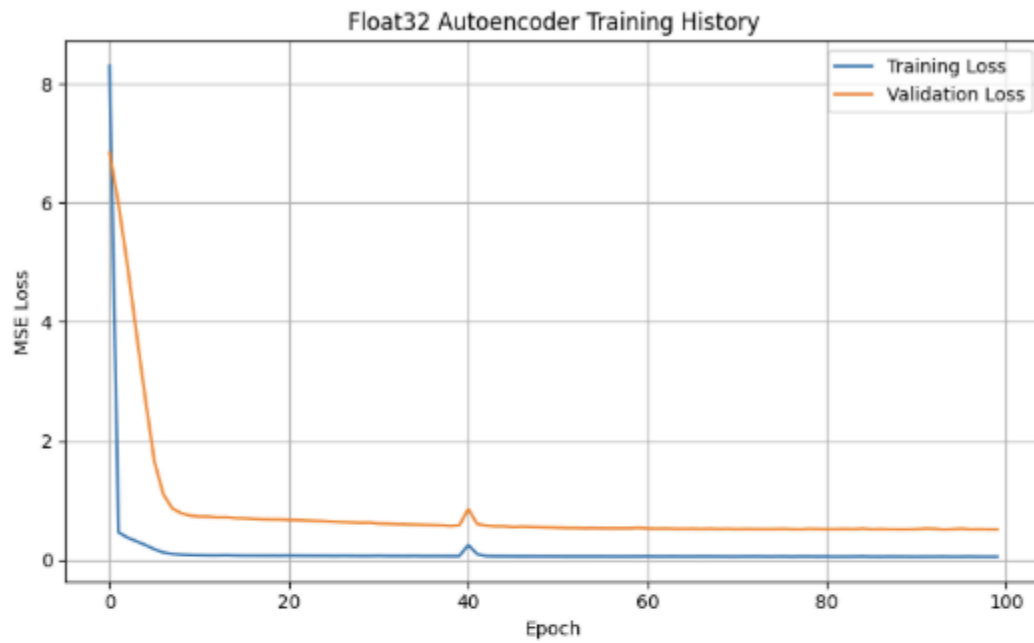
```
off: 0.000000
Anomaly prediction: 15.245161
Starting inferencing in 2 seconds...
Sampling...
Timing: DSP 44 ms, inference 618 us, anomaly 588 us, postprocessing 58 us
#Classification predictions:
nominal: 0.437508
off: 0.562508
Anomaly prediction: 16.024948
Starting inferencing in 2 seconds...
Sampling...
Timing: DSP 44 ms, inference 618 us, anomaly 587 us, postprocessing 49 us
#Classification predictions:
nominal: 0.437508
off: 0.562508
Anomaly prediction: 16.025192
Starting inferencing in 2 seconds...
Sampling...
Timing: DSP 44 ms, inference 592 us, anomaly 586 us, postprocessing 185 us
#Classification predictions:
nominal: 0.437508
off: 0.562508
Anomaly prediction: 16.025122
Starting inferencing in 2 seconds...
Sampling...
Timing: DSP 44 ms, inference 596 us, anomaly 524 us, postprocessing 968 us
#Classification predictions:
nominal: 0.402344
off: 0.597656
Anomaly prediction: 15.986572
Starting inferencing in 2 seconds...
Sampling...
Timing: DSP 44 ms, inference 593 us, anomaly 524 us, postprocessing 58 us
#Classification predictions:
nominal: 0.437508
off: 0.562508
Anomaly prediction: 16.018923
Starting inferencing in 2 seconds...
Sampling...
Timing: DSP 44 ms, inference 611 us, anomaly 525 us, postprocessing 49 us
#Classification predictions:
nominal: 0.437508
off: 0.562508
Anomaly prediction: 16.028332
Starting inferencing in 2 seconds...
Sampling...
Timing: DSP 44 ms, inference 592 us, anomaly 6 ms, postprocessing 49 us
#Classification predictions:
nominal: 0.886719
off: 0.113281
Anomaly prediction: 13.619226
Starting inferencing in 2 seconds...
Sampling...
Timing: DSP 44 ms, inference 613 us, anomaly 6 ms, postprocessing 49 us
#Classification predictions:
nominal: 0.710937
off: 0.289063
Anomaly prediction: 774.686934
Starting inferencing in 2 seconds...
Sampling...
Timing: DSP 44 ms, inference 611 us, anomaly 6 ms, postprocessing 49 us
#Classification predictions:
nominal: 0.898437
off: 0.101563
Anomaly prediction: 11.736388
Starting inferencing in 2 seconds...
#
```

6. During deployment, the model predicted "off" with ~56% confidence and "nominal" with ~44% confidence, with an anomaly score of ~16. Since the board wasn't connected to a fan, the accelerometer data didn't closely match either training class, which explains the low confidence split and high anomaly score. The classifier output shouldn't always be trusted on its own; when the confidence values are this close, it might be because the model is uncertain. As for the anomaly score, a high score indicates that the input data falls outside the learned normal clusters, which may cause the classification result to be unreliable.

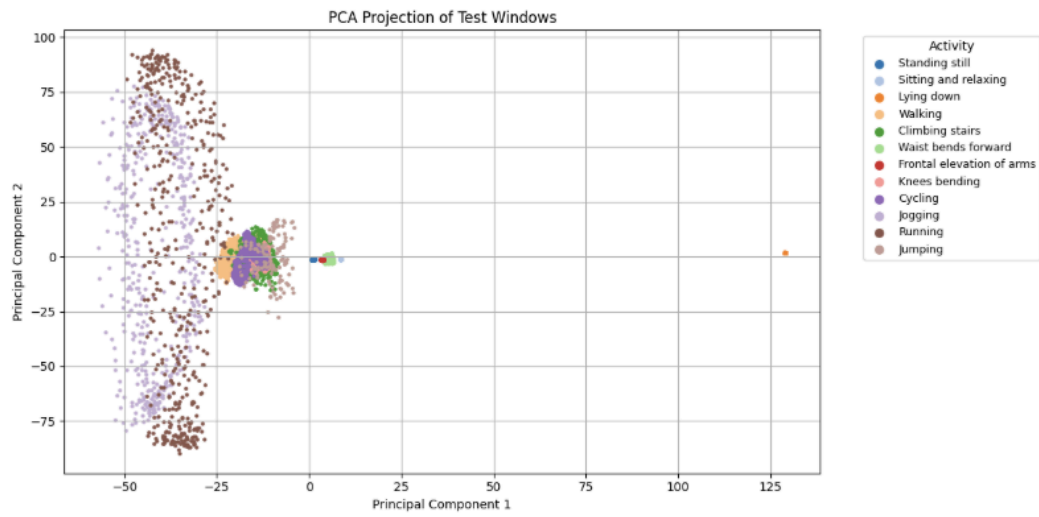
Section 2

Training and validation loss curves

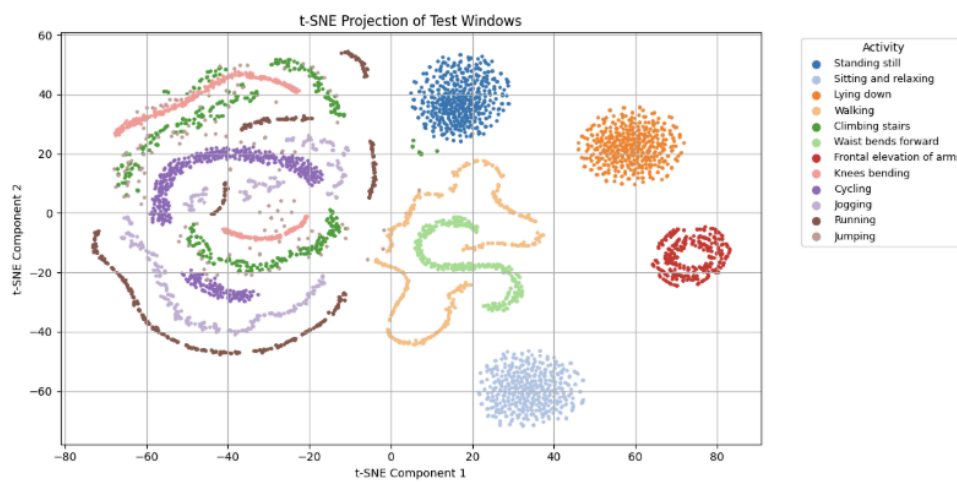
31/31 [=====] - 75 imgs/step - loss: 0.0043 - val_loss: 0.0119



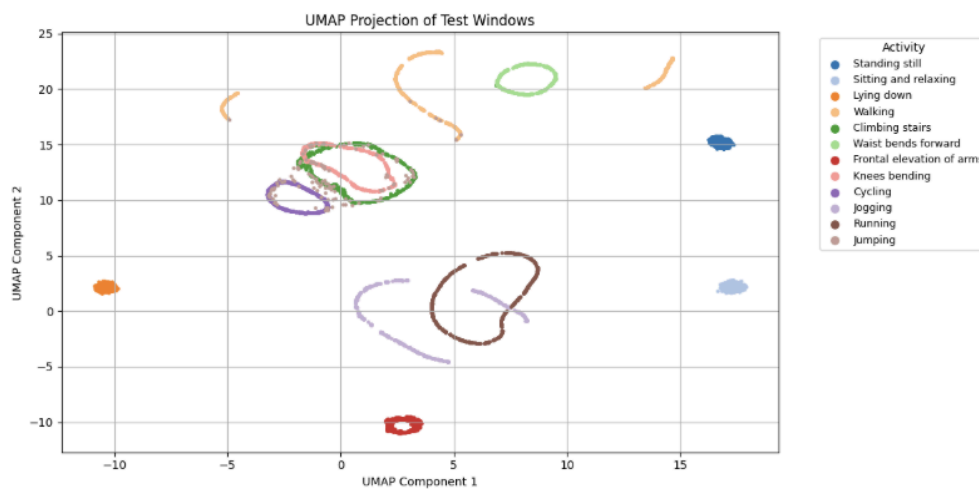
PCA visualization



t-SNE visualization



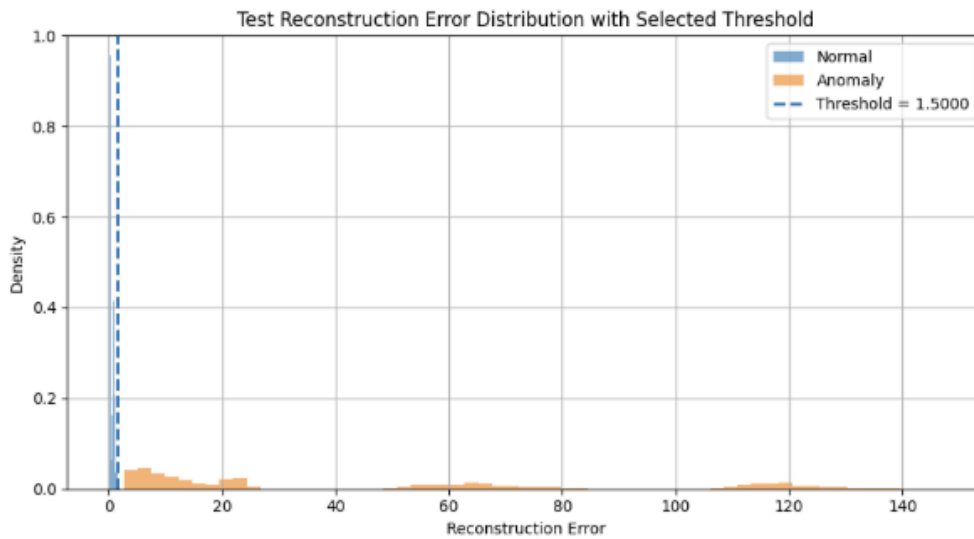
UMAP visualization



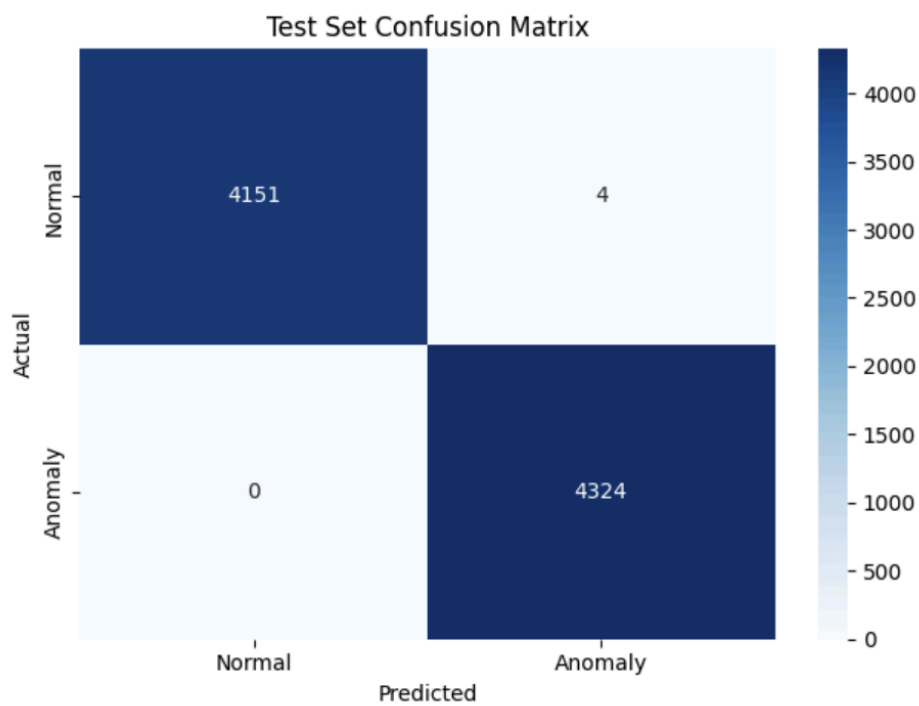
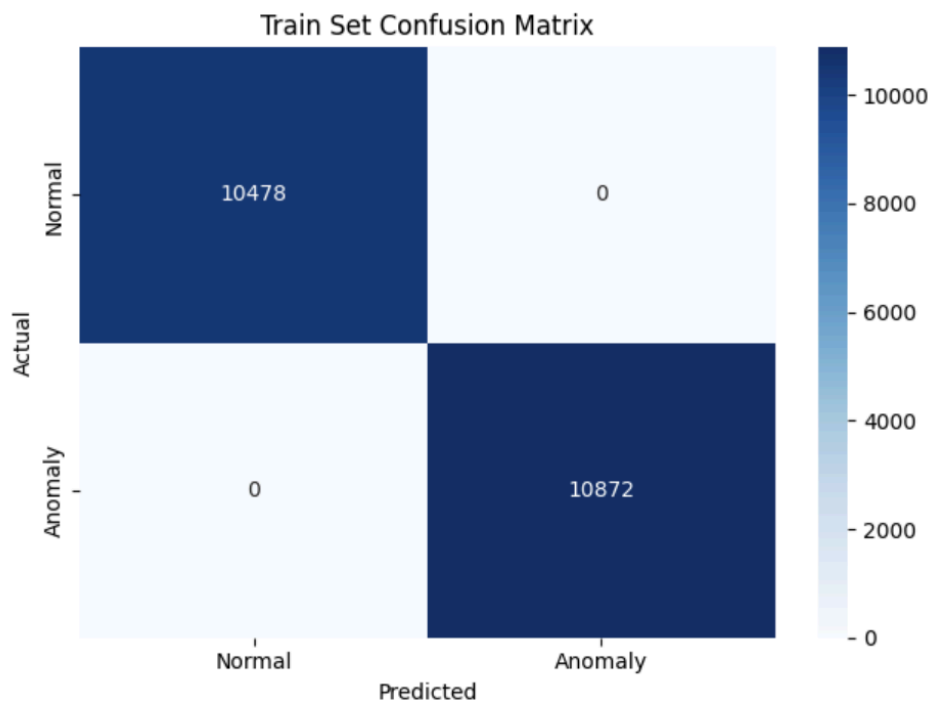
Reconstruction error distribution



Selected anomaly threshold



Confusion matrix



Classification report/evaluation metrics

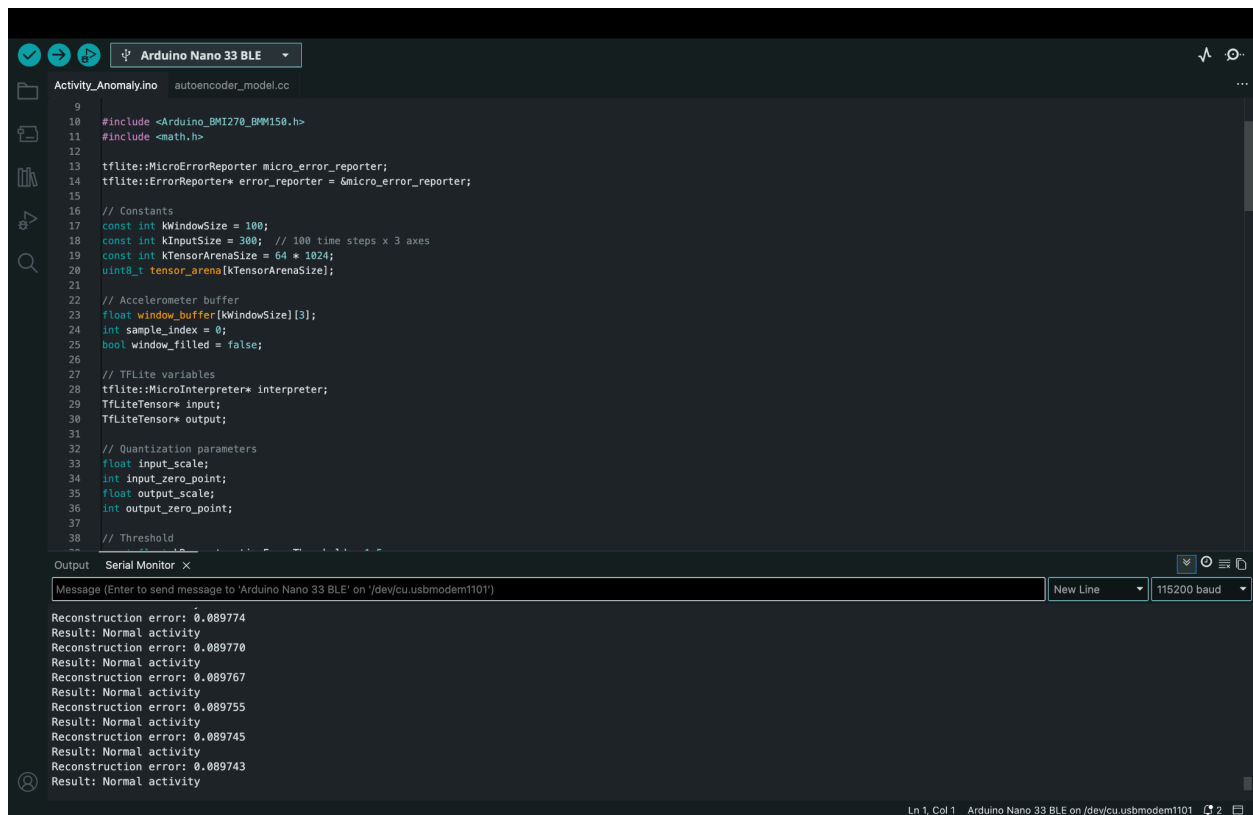
Train set evaluation

	precision	recall	f1-score	support
Normal (0)	1.00	1.00	1.00	10478
Anomaly (1)	1.00	1.00	1.00	10872
accuracy			1.00	21350
macro avg	1.00	1.00	1.00	21350
weighted avg	1.00	1.00	1.00	21350

Test set evaluation

	precision	recall	f1-score	support
Normal (0)	1.00	1.00	1.00	4155
Anomaly (1)	1.00	1.00	1.00	4324
accuracy			1.00	8479
macro avg	1.00	1.00	1.00	8479
weighted avg	1.00	1.00	1.00	8479

Serial monitor output



The screenshot shows an IDE window with the file 'Activity_Anomaly.ino' open. The code includes headers for Arduino_BMI270_BMM150 and math, defines TFLite error reporting, sets constants for window size, input size, and tensor arena, and declares variables for accelerometer buffer, TFLite interpreter, tensors, quantization parameters, and a threshold.

The serial monitor at the bottom shows the following output:

```
Reconstruction error: 0.089774
Result: Normal activity
Reconstruction error: 0.089770
Result: Normal activity
Reconstruction error: 0.089767
Result: Normal activity
Reconstruction error: 0.089755
Result: Normal activity
Reconstruction error: 0.089745
Result: Normal activity
Reconstruction error: 0.089743
Result: Normal activity
```

The status bar at the bottom indicates 'Ln 1, Col 1 Arduino Nano 33 BLE on /dev/cu.usbmodem1101'.